2024 AMC 10B Problem 22 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10B Problem 22. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10B Problem 22

Problem:
A group of 16 people will be partitioned into 4 indistinguishable 4-person committees. Each committee will have one chairperson and one secretary. The number of different ways to make these assignments can be written as $3^r M$, where r and M are positive integers and M is not divisible by 3 . What is r ?

Answer Choices:

A. 5

B. 6

C. 7

D. 8

E. 9

Solution:

The 16 people can be partitioned into the 4 committees, each of size 4, in

$$\frac{16!}{(4!)^5}$$

ways; four of the 4 ! factors come from permuting the members of the committees and one 4 ! factor comes from permuting the committees. Then there are 4^4 ways to choose the four chairpersons and 3^4 ways to choose the four secretaries. This gives a total of

$$\frac{16! \cdot 4^4 \cdot 3^4}{(4!)^5}$$

assignments. The numerator has 1 factor of 3 in each of $15, 12, 6$, and 3 ; it has 2 factors of 3 in 9 and 4 factors in 3^4, a total of 10 . The denominator has 5 factors of 3 . Thus $r = 10 - 5 = \textbf{(A)} \boxed{5}$.

OR

There are $16 \cdot 15$ ways to choose the chairperson and secretary of the first committee, $14 \cdot 13$ ways to choose the chairperson and secretary of the second committee, $12 \cdot 11$ ways to choose the chairperson and secretary of the third committee, and $10 \cdot 9$ ways to choose the chairperson and secretary of the fourth committee. There are then $\frac{8 \cdot 7}{2} = 28$ ways to choose the remaining members of the first committee, $\frac{6 \cdot 5}{2} = 15$ ways to choose the remaining members of the second committee, $\frac{4 \cdot 3}{2} = 6$ ways to choose the remaining members of the third committee, and $\frac{2 \cdot 1}{2} = 1$ way to choose the remaining members of the fourth committee. There are 4! ways to account for the fact that the committees are indistinguishable. This gives a total of

$$\frac{16 \cdot 15 \cdot 14 \cdot 13 \cdot 12 \cdot 11 \cdot 10 \cdot 9 \cdot 28 \cdot 15 \cdot 6 \cdot 1}{4!}$$

assignments. There are $1 + 1 + 2 + 1 + 1 = 6$ factors of 3 in the numerator and 1 factor of 3 in the denominator, so $r = 6 - 1 = \textbf{(A)} \boxed{5}$.

Note: The number of ways to make the assignments is $54,486,432,000 = 3^5 \cdot 224,224,000$.